

## 2 Shared root zone foundation

Avocado feeder roots are concentrated near the surface and need a workable balance of water and air.

### Soil is a four part system

| Part                     | Main job                                                                        | When out of balance                                                                                |
|--------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Mineral particles</b> | Provide the sand silt and clay framework and many nutrient-holding sites.       | Texture may drain too quickly or hold water too long.                                              |
| <b>Organic matter</b>    | Feeds soil life helps aggregation holds water and nutrients and buffers change. | Low inputs can weaken structure and biological activity; poor-quality inputs can add salts.        |
| <b>Water</b>             | Carries dissolved nutrients to roots and replaces crop water use.               | Too little causes drought and salt concentration; too much displaces air and favors root problems. |
| <b>Air</b>               | Supplies oxygen for root and microbial respiration.                             | Poor drainage or compaction limits oxygen and slows nutrient uptake.                               |

### Texture and structure are different

- **Texture is the mix of sand silt and clay.** It changes slowly and strongly affects water storage, drainage, and nutrient retention.
- **Structure is how particles are arranged into crumbs and pores.** It can improve or decline with organic inputs, roots, traffic, sodium, wetting, and disturbance.
- **Sandy soil usually stores less water and has a lower ability to hold potassium.** It often needs closer attention to irrigation timing and nutrient losses.
- **Clayey soil can store more water and nutrients, but structure matters.** Compaction, sodium, or poor drainage can leave roots wet and short of oxygen.

### Soil quality functions to inspect

| Function                    | Good field sign                                                     | Concern                                                                        |
|-----------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Infiltration</b>         | Water enters without long ponding or runoff.                        | Surface sealing compaction sodium or an application rate faster than intake.   |
| <b>Water availability</b>   | Moisture reaches active roots and is renewed before severe stress.  | Dry pockets short run times shallow wetting or roots forced into saltier soil. |
| <b>Drainage</b>             | Excess water can leave the active root zone.                        | Persistent wetness perched water restrictive layers or saturated low spots.    |
| <b>Aeration and rooting</b> | Pores refill with air and fine feeder roots occupy the wetted zone. | Water-filled pores low oxygen and black brittle sparse or shallow roots.       |
| <b>Leaching</b>             | Enough water moves salts below roots and has a drainage outlet.     | Salt remains in the root zone or added water creates waterlogging.             |